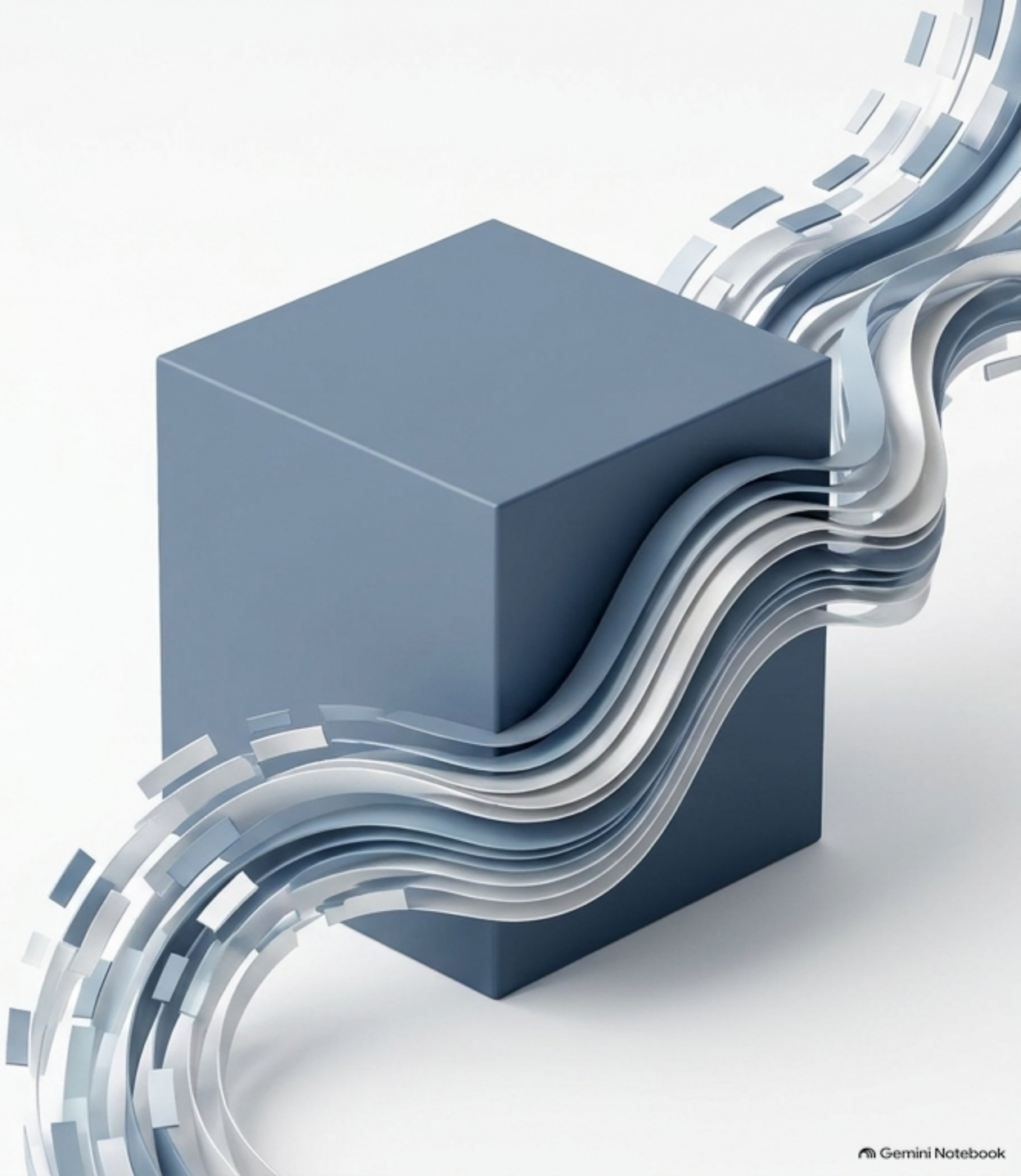
Beyond the Prompt

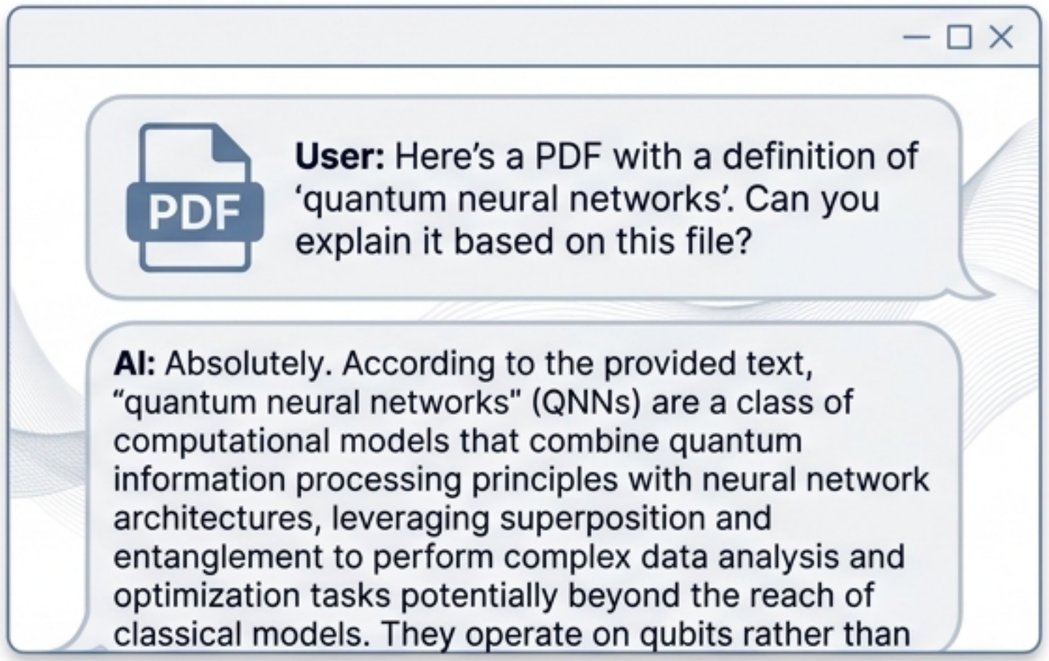
Demystifying Training, Inference, and RAG in Generative AI.



Gemini Notebook

The Illusion of Learning

Temporary Inputs



Structural Permanence



When a model uses a definition you just inserted into the prompt, did its internal parameters actually update?

Fluent writing does not mean structural learning.

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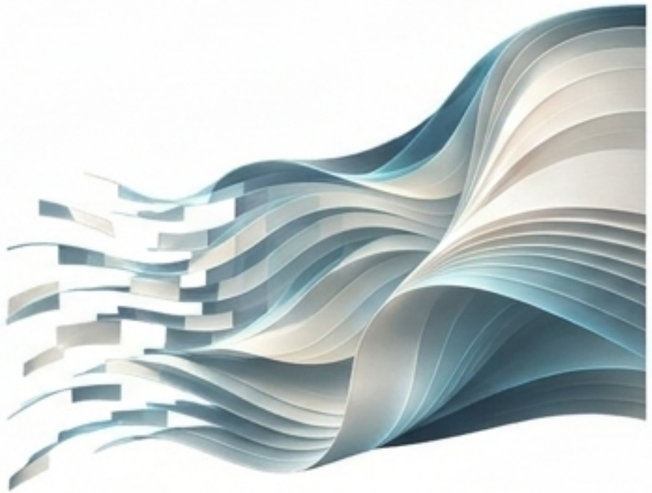
Two Distinct Interventions

Structural Level



Parameters: The learned numerical values inside the model. Changing these requires a completely new optimization and training process.

Input Level

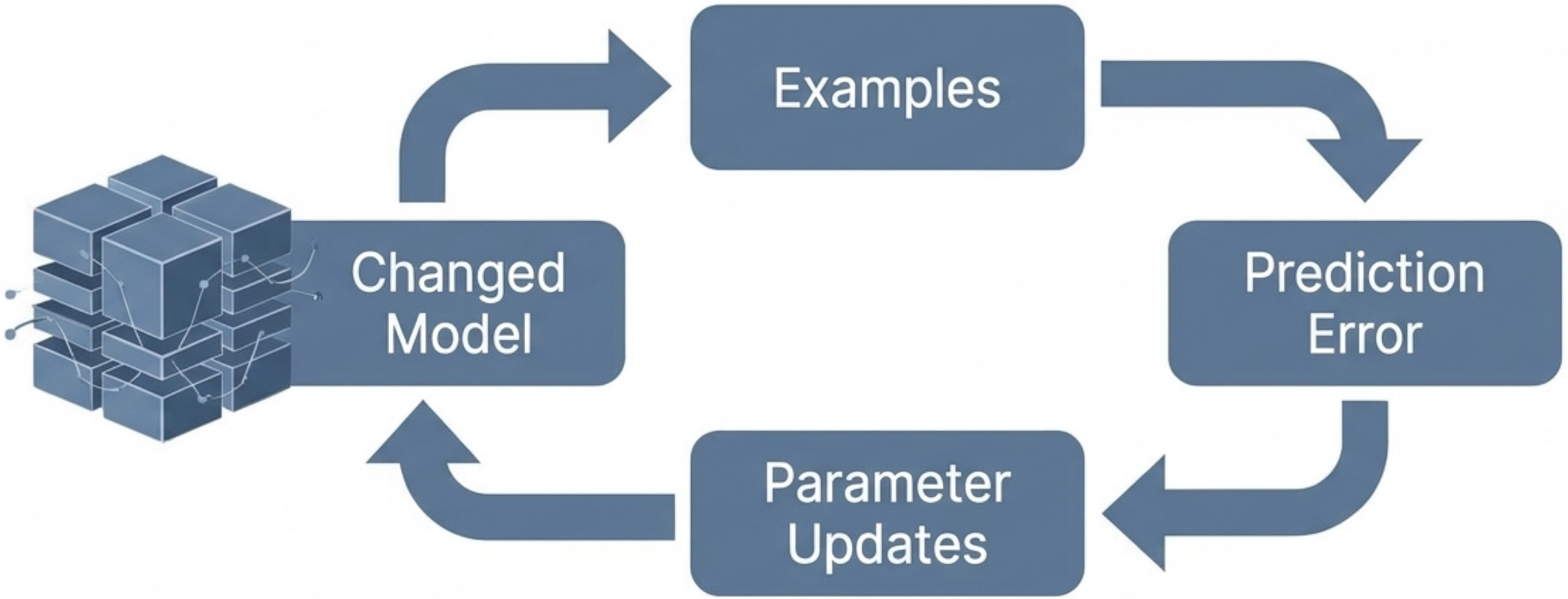


Context: The sequence of information made available to the current calculation. Changing this only requires a new prompt.

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Model Training: Rewiring the Inside

Pretraining compares predicted distributions with actual text, updating the **parameters** to reduce aggregate error.



The text supplies the learning signal; optimization updates the actual values.

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Ordinary Inference: Fixed Parameters, New Data

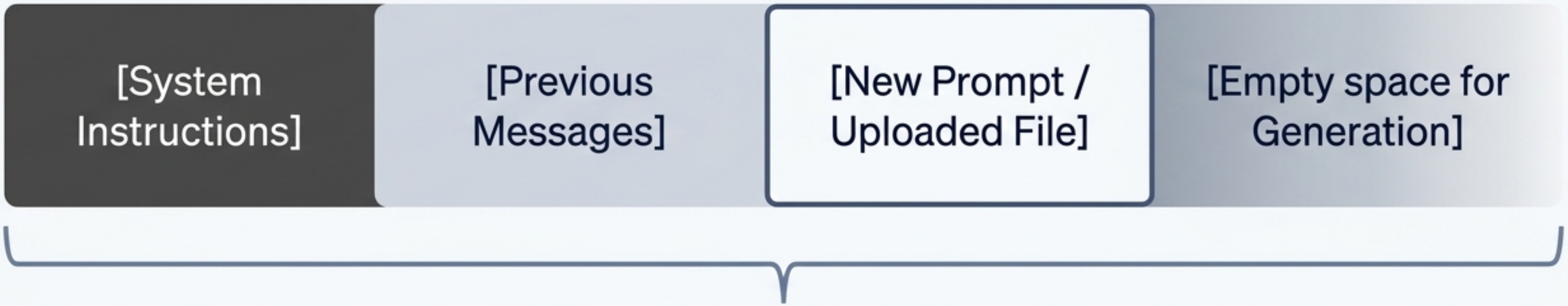
The model runs calculations with numerical values that are already available.
The model itself remains fixed.



A new message changes the information available to the current calculation, not the model itself.

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Inside the Context Window



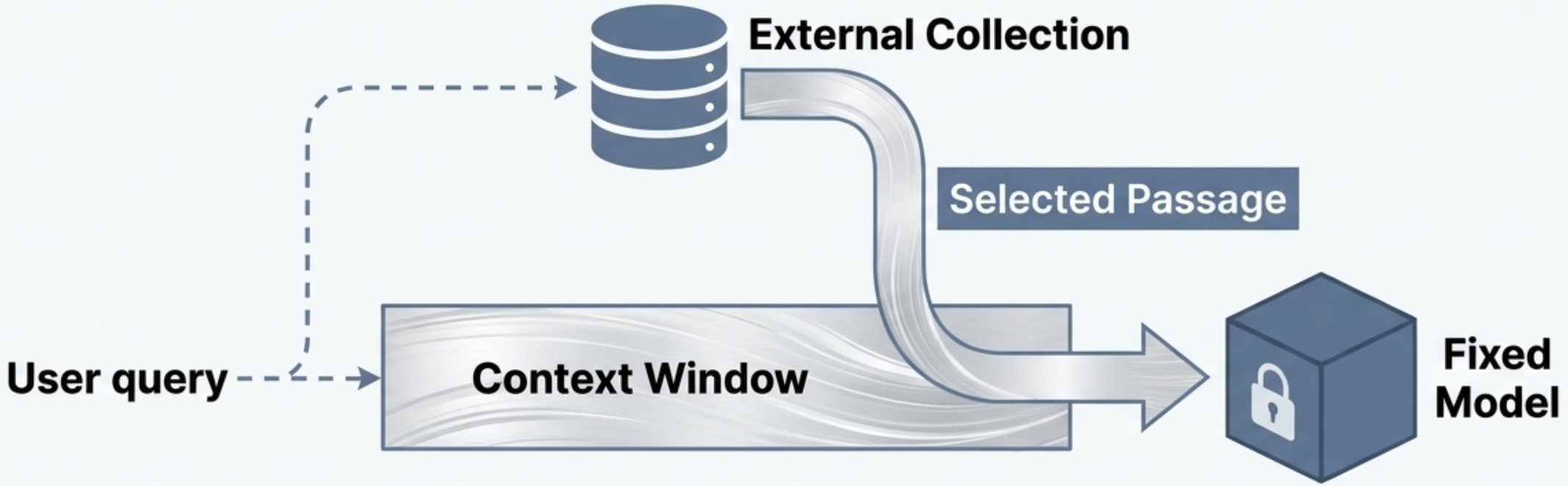
The sequence of tokens made available to the current inference. It is measured in tokens, not words. It dictates exactly what the fixed model ‘sees’ at this exact moment.

Once the session ends, this context evaporates entirely.

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RAG: Automating the Context

A separate component searches external documents and makes selected passages available to the generation process.



RAG updates the information available in context without changing the model's weights.

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	Training	Inference	RAG
Inputs	Examples	Input + Fixed Model	Question + Retrieved Passages
Core Action	Parameter Updates	Model Calculations	Context + Model Calculations
Result	Changed Model	Generated Answer	Generated Answer
What Changes	Model Parameters	Current Information	External Context Supplied

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Concept Check

An assistant app is provided with a proprietary 500-page company manual via a database connection. It now answers employee questions perfectly based on that manual.

Explain to a colleague why it is technically incorrect to say the model was ‘fine-tuned’ on the manual. What actually happened?

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